EXPLORATORY DATA ANALYSIS

CHAPTER 7

Descriptive Statistics: Beyond the Basics

A deeper quantitative toolkit for summarizing, comparing, and interpreting data

Focus: robust statistics, trimmed and winsorized means, weighted averages, relative variability, quantiles, standard errors, and the limits of summary statistics.

Scope: descriptive analysis of observed data—not inferential hypothesis testing, feature scaling, feature engineering, or model training.

Learning Objectives

- Explain why no single statistic adequately describes a distribution.
- Distinguish population and sample descriptive formulas and understand Pandas defaults.
- Calculate and interpret trimmed means, winsorized means, weighted means, and the coefficient of variation.
- Understand quantiles, percentiles, IQR, and quantile-based summaries.
- Distinguish standard deviation from standard error.
- Compare groups using center, spread, sample size, and distribution shape.
- Produce defensible summaries with clear units, denominators, and caveats.

Review of Earlier Chapters

Chapter 1 — What Is Exploratory Data Analysis?

EDA is an iterative investigation, not merely a set of plots. Quantitative summaries and visual inspection complement each other. A statistic describes an aspect of observed data; it does not automatically explain why it exists.

Chapter 2 — Understanding the Dataset

Establish observational unit, variable meaning, granularity, and data-generating process before summarizing. A customer-level mean is not interchangeable with a transaction-level mean.

Chapter 3 — Data Quality Assessment

Completeness, validity, consistency, uniqueness, accuracy, and timeliness affect every summary. Duplicates, placeholders, impossible values, and inconsistent units can distort means, spread, and counts.

Chapter 4 — Exploring Numerical Variables

Mean, median, mode, range, variance, SD, quartiles, IQR, skewness, and outlier investigation form the foundation. This chapter explains when these familiar summaries can mislead.

Chapter 5 — Exploring Categorical Variables

Counts, proportions, denominators, cardinality, rare categories, and ordered categories matter when comparing groups. Group summaries should include sample sizes and context.

Chapter 6 — Exploring Date and Time Variables

Temporal coverage, event frequency, durations, and inter-event gaps create numerical quantities that can be summarized with the tools in this chapter.

Review of Chapter 6 — Exploring Date and Time Variables

- Temporal coverage: minimum and maximum timestamps, observed span, distinct dates, and missing periods.
- Describe durations and inter-event gaps using center, spread, quantiles, and units.
- Right-skewed waiting times can make the mean much larger than the median.
- Sort events within each entity before calculating consecutive gaps.
- A zero-count day may indicate no events or a collection outage; context is required.

Chapter 6 Assignment and Project — Discussion Guide

No submitted answers or notebook accompanied this request, so this is an expected-reasoning guide rather than a grading report.

- Calculate gaps only after sorting; treat the first gap per entity as unavailable, not zero.
- Investigate large gaps against expected cadence, active periods, and collection logs.
- For mixed date formats or time zones, preserve raw values and audit parsing failures.
- A temporal profiler should report missingness, parsing failures, valid range, distinct timestamps, and time-zone assumptions.

Expected project learning: derive auditable durations and frequencies from timestamps, summarize them with explicit units, and document source assumptions.

Preview of This Chapter

The central question is not merely which formula can be calculated, but which summary best represents this distribution—and what information that summary hides.

1. Why Go Beyond the Mean and Standard Deviation?

A statistic compresses a distribution. Datasets can share a mean while differing in clusters, tails, gaps, and outliers. A useful summary combines location, spread, shape, scale, and context.

Dataset	Values	Mean
A	4, 5, 5, 5, 6	5
B	1, 3, 5, 7, 9	5
C	0, 0, 5, 10, 10	5

The mean alone does not reveal concentration, multimodality, skewness, or unusual observations. Pair it with suitable spread measures and visual checks.

2. Population Versus Sample Variance

For values x■,…,x■, population variance divides squared deviations from the population mean μ by N: **σ² = Σ(x■ − μ)² / N**. Sample variance commonly divides squared deviations from the sample mean x■ by n − 1: **s² = Σ(x■ − x■)² / (n − 1)**.

The n−1 denominator is Bessel's correction. Under the usual random-sampling framework it makes sample variance an unbiased estimator of population variance. It is not a universal instruction to always divide by n−1; match the formula to the quantity being described.

Pandas uses sample variance and sample SD by default for a Series (`ddof=1`). Use `ddof=0` for population-form calculations on the observed values.

```
x = pd.Series([2, 4, 6, 8])
sample_var = x.var()          # ddof=1
population_var = x.var(ddof=0)
sample_sd = x.std()
population_sd = x.std(ddof=0)
```

State the convention when reporting variance or standard deviation.

3. Quantiles and Percentiles

A p-quantile is a value with approximately proportion p of observations at or below it. The median is the 0.50 quantile; Q1 and Q3 are the 0.25 and 0.75 quantiles. Percentiles express the same idea on a 0–100 scale.

```
x = pd.Series([3, 4, 4, 5, 7, 8, 10, 12, 20])
q = x.quantile([.05, .25, .50, .75, .90, .95])
print(q)
```

A requested quantile may fall between observed values. Software uses an interpolation convention, so results can differ for small or discrete samples.

```
for method in ["linear", "lower", "higher", "nearest", "midpoint"]:
    print(method, x.quantile(.25, interpolation=method))
```

Record the interpolation method when exact reproducibility matters. Quantiles can be unstable with small samples, many ties, or rounded measurements.

4. IQR and Quantile-Based Spread

The interquartile range is **IQR = Q3 − Q1**. It measures the width of the middle 50% and is less sensitive to extreme endpoints than range or SD. It does not describe the tails or the arrangement of values inside the middle half.

```
q1, med, q3 = x.quantile([.25, .50, .75])
iqr = q3 - q1
print(q1, med, q3, iqr)
```

The five-number summary is minimum, Q1, median, Q3, maximum. Because minimum and maximum are sensitive to extremes, a percentile summary such as P05, P25, P50, P75, P95 can be more informative.

5. Robust Measures of Location

Median

The median is the 50th percentile. It is often a useful typical value for skewed distributions or data containing extreme observations.

Trimmed mean

A trimmed mean removes a specified proportion from each tail before calculating the mean. A 10% trim removes approximately the lowest 10% and highest 10%.

```
from scipy import stats
values = pd.Series([12, 13, 13, 14, 15, 15, 16, 17, 18, 120])
ordinary_mean = values.mean()
median_value = values.median()
trimmed_mean = stats.trim_mean(values, proportiontocut=.10)
```

Trimming is a deliberate summary rule, not permission to hide inconvenient observations. State the trim proportion, retain the ordinary summary for transparency, and be cautious with small samples.

Winsorized mean

A winsorized mean replaces tail observations with selected cutoff values rather than removing them. It retains the sample size while limiting the influence of extremes.

```
from scipy.stats.mstats import winsorize
winsorized = winsorize(values, limits=(.10, .10))
winsorized_mean = float(winsorized.mean())
```

Trimming excludes tail values from the calculation; winsorizing caps them. Neither is automatically correct. Use them as sensitivity summaries and investigate why extremes occurred.

6. Weighted Means

An ordinary mean gives each observation equal weight. A weighted mean is **Σ(w■x■) / Σw■**. Weights must have a defensible meaning—for example, transaction counts when combining store-level transaction averages.

```
scores = pd.Series([70, 80, 90])
weights = pd.Series([1, 2, 3])
weighted = np.average(scores, weights=weights)
ordinary = scores.mean()
```

Weighted means answer different questions, not necessarily better questions. If Store A has 2 transactions averaging 100 and Store B has 100 transactions averaging 50, the unweighted mean of store means is 75, while the transaction-weighted mean is (2×100 + 100×50)/102 ≈ 50.98. The latter answers the average across all transactions; the former averages the two store means equally.

7. Relative Variability: Coefficient of Variation

The coefficient of variation is **CV = standard deviation / mean**, often multiplied by 100%. It expresses variability relative to the mean and is dimensionless.

```
x = pd.Series([90, 100, 110, 100, 100])
cv = x.std() / x.mean()
cv_percent = 100 * cv
```

CV is most interpretable for ratio-scale measurements with a meaningful zero and a positive mean. It can be unstable when the mean is near zero, negative, or when the scale lacks a meaningful zero. Do not use it mechanically for Celsius temperature, arbitrary scores, or centered variables.

8. Standard Deviation Versus Standard Error

SD describes spread among individual observations. Standard error describes the variability of a statistic across repeated samples under a sampling model. For the sample mean, the conventional estimated SE is **SE(x■) = s / √n**.

```
x = pd.Series([8, 9, 10, 10, 11, 12, 15])
sd = x.std()
se_mean = sd / np.sqrt(x.count())
```

A large dataset can have a small SE while individual values remain highly variable. Do not report SE in place of SD when the goal is to describe the spread of the data.

9. Skewness, Outliers, and Choosing Summaries

Situation	Useful summaries	Caution
Roughly symmetric	Mean, SD, median, IQR	Check clusters and tails
Right-skewed	Median, IQR, P90/P95, mean	Mean may be pulled upward
Left-skewed	Median, IQR, lower quantiles	Mean may be pulled downward
Extreme values	Median, IQR, quantiles, mean sensitivity	Investigate, do not auto-delete
Multiple clusters	Group-specific summaries	One center may describe no cluster

A robust summary does not remove the need to investigate unusual observations. They may be errors, valid rare events, or evidence of different subpopulations.

10. Comparing Groups

For group comparisons, report n, mean, median, SD, Q1, Q3, and useful endpoints. A group with n=3 should not be interpreted like one with n=3,000. Missing values may cause different statistics to use different effective sample sizes.

```
group_summary = (
    df.groupby("segment")["amount"]
      .agg(
          n="count", mean="mean", median="median",
          sd="std",
          q1=lambda s: s.quantile(.25),
          q3=lambda s: s.quantile(.75),
          minimum="min", maximum="max"
      )
)
```

Group differences are descriptive, not automatically causal. Selection, time period, confounding, and measurement differences may explain apparent gaps.

11. Missing Values and Denominators

Pandas descriptive methods commonly skip missing values. A mean may therefore use fewer observations than the dataset row count. Report valid n and missing n.

```
s = pd.Series([10, 12, np.nan, 16, 100])
summary = {
    "rows": len(s),
    "valid_n": int(s.count()),
    "missing_n": int(s.isna().sum()),
    "mean": s.mean(),
    "median": s.median(),
    "sd": s.std()
}
```

When comparing groups or variables, check that summaries use comparable populations. Different missingness patterns can change the set of observations being summarized.

12. A Reusable Numerical Summary

This starter function reports complementary summaries. For small samples, some quantiles and skewness estimates may be unstable; document interpolation and ddof.

```
def describe_numeric(series):
    s = pd.to_numeric(series, errors="coerce")
    v = s.dropna()
    if v.empty:
        return {"rows": len(s), "valid_n": 0,
                "missing_n": int(s.isna().sum())}
    q = v.quantile([.05, .25, .50, .75, .90, .95])
    return {
        "rows": len(s), "valid_n": len(v),
        "missing_n": int(s.isna().sum()),
        "mean": v.mean(), "median": v.median(),
        "sample_sd": v.std(ddof=1),
        "minimum": v.min(), "p05": q.loc[.05],
        "q1": q.loc[.25], "q3": q.loc[.75],
        "p90": q.loc[.90], "p95": q.loc[.95],
        "maximum": v.max(), "iqr": q.loc[.75]-q.loc[.25],
        "skewness": v.skew()
    }
```

A professional toolkit should also document units, validity rules, quantile interpolation, and whether ddof=0 or 1 is used.

13. Visual Checks

Although this chapter emphasizes quantitative EDA, summary tables should be checked against distribution plots. Use Matplotlib's object-oriented API.

```
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
axes[0].hist(values, bins=10, edgecolor="black")
axes[0].set_title("Observed Values")
axes[0].set_xlabel("Value")
axes[1].boxplot(values)
axes[1].set_title("Boxplot")
axes[1].set_ylabel("Value")
fig.tight_layout()
```

Plots may reveal clusters, gaps, skew, and outliers hidden by a compact table. They complement rather than replace quantitative summaries.

14. Common Mistakes

- Reporting a mean without n, units, or context.
- Treating sample and population variance as interchangeable.
- Assuming the median or trimmed mean is always superior.
- Using weights without explaining their meaning.
- Using CV near a zero mean or on a scale without a meaningful zero.
- Confusing SE with the spread of individual observations.
- Reporting quantiles without considering interpolation and sample size.
- Comparing group means while ignoring group sizes, spread, and missingness.
- Deleting extreme values without investigating them.
- Treating descriptive group differences as causal effects.

Chapter Summary

- No single statistic reveals a complete distribution.
- Population and sample variance use different denominators; Pandas defaults to ddof=1.
- Quantiles and IQR describe distribution position and robust spread.
- Trimmed and winsorized means handle tails differently.
- Weighted means are appropriate only when weights match the target quantity.
- CV requires a meaningful ratio-scale zero and a suitable positive mean.
- SD describes individual spread; SE describes sampling variability of a statistic.
- Group summaries need sample size, center, spread, units, and caveats.

Assignment

Part A — Conceptual Questions

- Why can two datasets have the same mean but very different distributions?
- When is the median more informative than the mean?
- Explain sample versus population variance and Pandas' default ddof.
- Compare trimmed and winsorized means.
- What question does a weighted mean answer?
- Why can CV be inappropriate or unstable?
- Distinguish SD from SE precisely.
- Why should group summaries include n, median, and spread?

Part B — Manual Quantitative Analysis

```
x = [4, 5, 5, 6, 7, 8, 9, 10, 12, 34]
```

- Calculate mean and median.
- Calculate range, Q1, Q3, and IQR using Pandas default linear interpolation.
- Calculate sample SD and population SD.
- Explain how 34 affects mean and SD.
- Choose suitable summaries for typical value and spread, and justify your choice.

Part C — Weighted Means

Branch	Transactions	Mean amount
A	5	120
B	95	80

- Calculate the unweighted average of branch means.
- Calculate the transaction-weighted average.
- Explain what each represents.
- Which answers the average transaction amount across all transactions?

Part D — Coding Assignment

- Choose a numerical column from a real or practice dataset.
- Report rows, valid n, missing n, mean, median, sample SD, population SD, min, max, Q1, Q3, IQR, P05, P90, and P95.
- Calculate trimmed and winsorized means where sample size makes them sensible.
- Calculate CV only if mathematically and substantively meaningful; otherwise explain why not.
- Create group summaries with n, mean, median, SD, Q1, and Q3.
- Compare mean and median and check the distribution visually.
- Write a short report with units, denominators, assumptions, and limitations.

Coding Project — Descriptive Statistics Toolkit

Extend the cumulative EDA master notebook with a reusable quantitative-summary module. Keep it integrated with earlier numerical, categorical, and temporal EDA sections.

- Implement describe_numeric() with valid/missing n, mean, median, sample and population SD, endpoints, quantiles, IQR, and skewness.
- Implement trimmed-mean and winsorized-mean comparisons with documented tail proportions.
- Implement weighted_mean() with nonnegative-weight validation and explicit handling of zero total weight.
- Implement coefficient_of_variation() with checks for unsuitable means and scales.
- Implement grouped_numeric_summary() with n and multiple center/spread measures.
- Make quantile interpolation and ddof choices explicit.
- Test on symmetric, skewed, outlier-containing, small, and missing-value datasets.
- Compare summaries with at least two Matplotlib object-oriented visual diagnostics.
- Write interpretations explaining which statistics are most informative and why.

Expected reusable components: describe_numeric(), trimmed/winsorized comparison, weighted_mean(), coefficient_of_variation(), and grouped_numeric_summary(). Add docstrings, input checks, and clear return values.

Deliverables: updated cumulative master notebook, concise quantitative findings, and exported diagnostic plots.

Preview of Next Chapter

Chapter 8 — Understanding Distributions

We'll study distributions as complete objects: frequency and density, empirical cumulative distribution functions, probability mass and density ideas, multimodality, tails, and comparison across groups. The emphasis will be on reading and reasoning about distribution shape—not merely choosing a summary statistic.